

The Cascade Curvature Operator under Refinement: Schur–Complement Identity and Energy Intertwining

A predecessor to the cascade-closure mechanism paper

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Abstract

We isolate a clean abstract model — a scalar vertex-sector graph Laplacian under pure edge-midpoint subdivision with degree-2 midpoints — and prove a Schur-complement halving identity in this model that lifts the source’s coboundary factor- $\frac{1}{2}$ relation $p^1 d_{n+1} = \frac{1}{2} d_n p^0$ from [2] to the curvature level. With the scaled vertex Laplacian $\tilde{A}_n := 2^n A_n^{\text{vertex}}$, the Schur complement under the (boundary, interior) partition equals the previous-level operator exactly: $\text{Schur}(\tilde{A}_{n+1}) = \tilde{A}_n$. The identity is verified analytically and numerically on two concrete instances of the abstract model (single edge with one-midpoint subdivision; triangle with edge-midpoint subdivision) and yields the cascade-mechanism residual-monotonicity hypothesis $\tilde{R}_n(p^0 \psi) \leq \tilde{R}_{n+1}(\psi)$ for every $\psi \in X_{n+1}^0$, with equality if (but not in general only if) ψ is the harmonic extension of $p^0 \psi$. Strict operator-level refinement-compatibility $p^0 \tilde{A}_{n+1} = \tilde{A}_n p^0$ remains false; the defect operator vanishes on harmonic extensions and is generically non-zero on the level- n -supported sector $\{\psi : \psi_I = 0\}$. We *do not* extend the result to the full Schläfli-refinement P3/P4 tower of [3, 2] (which adds face-centroids, midpoint-centroid edges, and F -valued chain spaces with antisymmetric oriented-edge convention); the gap between this paper’s abstract scalar pure-midpoint model and that fuller setting is named explicitly in §1.1 as the remaining upstream task. The result discharges cascade-mechanism’s obligation (O3) for the abstract model and partially discharges (O2) at the boundary-energy / Schur level for the abstract model, recasting the cascade-mechanism worked-invocation status table accordingly.

Scope

- This is a narrow technical paper on an *abstract scalar model* of refinement (vertex-sector graph Laplacian, pure edge-midpoint subdivision with degree-2 midpoints), not the full Schläfli-refinement P3/P4 tower of [3, 2] (see §1.1). It contains no new mechanism, no programme framing, no consciousness or cosmology claim. It is the source-side companion to [1], applied to the abstract model.
- Within the abstract scalar model: (O0) closed; (O3) closed via the Schur halving identity (Theorem 2.1, Theorem 3.1); (O2) closed at the Schur / boundary-energy level only (Proposition 4.1); (O1) recorded but not nontrivially discharged (Proposition 5.1).
- Lifting these results to the full P3/P4 tower (centroids, midpoint-centroid extra edges, F -valued X_n^0 and antisymmetric X_n^1 chain spaces) is named as the remaining upstream task and is not attempted here.

1 Introduction and source-state recap

The cascade-closure mechanism paper [1] defines cascade closure as a rung-indexed closure-residual / projection-stack event, proves a conditional residual-zero / fixed-point propagation proposition, and supplies a consumer API listing four obligations that successor papers must discharge to invoke cascade closure as a load-bearing premise:

- (O0) Standing analytic / admissibility hypotheses.
- (O1) Top-rung fixed point and convergence.
- (O2) Flow intertwining $\pi_{k+1,k}^\# \circ \Psi_{k+1}^t = \Psi_k^t \circ \pi_{k+1,k}^\#$.
- (O3) Residual monotonicity $R_k(\pi_{k+1,k}^\# \psi) \leq R_{k+1}(\psi)$.

[1] §3.5 records the per-obligation status in the imported P3/P4 finite-level tower of [3, 2]: (O0) is satisfied at the finite-dimensional level; (O1)–(O3) are open in the source state, with (O2) closed only for the source’s mass-only operator $L_n = -\gamma C_n$ at $\alpha = \beta = 0$, not for the curvature/residual operator A_n that the cascade-mechanism flow naturally uses. [1] §3.5 names the minimal upstream upgrade required and observes that the role of [1] is to make the obligations visible as the programme’s actual dependency gate, not to discharge them.

The present paper is a partial source-side discharge, carried out in a deliberately scaled-down setting: an *abstract scalar pure-midpoint refinement model* (Definition 1.1 below) that retains the load-bearing coboundary refinement relation of the imported tower while dropping the centroids, midpoint-centroid edges, and F -valued / antisymmetric oriented-edge structure of the full Schläfli refinement. In this abstract model, with the curvature operator A_n rather than the source’s mass-only L_n , we supply a single source-side identity and exhibit its consequences for (O2) and (O3). The lift to the full P3/P4 tower of [3, 2] is named explicitly in §1.1 as hypotheses (L1)–(L3) and is not attempted here. We do not redefine the cascade-closure mechanism, do not modify [1]’s consumer API, and do not introduce any programme-level claim.

1.1 Scope: abstract scalar pure-midpoint model vs. full P3/P4 tower

The actual source refinement of [3] (`cascade-refinement-spaces.tex` lines 273–287) is the *Schläfli refinement*: at each level n it adds, alongside the new edge-midpoints $M_n^\bullet$, a separate set of face-centroids $C_n^\bullet$ (one per 2-face of the underlying 4-polytope), together with extra level- $(n+1)$ edges connecting midpoints to centroids and centroids to other midpoints in a $W^\bullet$ -equivariant pattern. The chain spaces in the source are also F -valued (`cascade-refinement-spaces.tex` lines 464–493): $X_n^{0,\bullet}$ is F^0 -valued on $V(G_n^\bullet)$ and $X_n^{1,\bullet}$ is F^1 -valued and antisymmetric on *oriented* edges, with edge-parent map $\pi_{n+1,n}^E$ taking the value $\perp$ on the new midpoint-centroid and centroid-midpoint edges.

The present paper does *not* work in this fuller setting. We work in a deliberate *abstract scalar pure-midpoint model* defined as follows:

Definition 1.1 (Abstract scalar pure-midpoint refinement model). A pair of finite undirected graphs $G_n = (V_n, E_n)$ and $G_{n+1} = (V_{n+1}, E_{n+1})$ together with bonding maps is in the abstract scalar pure-midpoint model when:

- (M1) $V_{n+1} = V_n \sqcup M_n$, where M_n contains exactly one new *midpoint* vertex m_e per edge $e \in E_n$, and no other new vertices (no centroids);
- (M2) $E_{n+1} = \{(u, m_e), (m_e, v) : e = (u, v) \in E_n\}$, i.e. every level- $(n+1)$ edge is a child of exactly one level- n edge (no extra midpoint–midpoint edges);
- (M3) the chain spaces are scalar: $X_n^0 = \mathbb{R}^{V_n}$, $X_n^1 = \mathbb{R}^{E_n}$, with standard ℓ^2 inner products and *unoriented* edge convention;

- (M4) bonding maps are vertex-restriction $p_{n+1,n}^0 : X_{n+1}^0 \rightarrow X_n^0$ along the inclusion $V_n \subset V_{n+1}$ and parent-fibre average $p_{n+1,n}^1 : X_{n+1}^1 \rightarrow X_n^1$ by factor $\frac{1}{2}$ on each of the two children of every parent edge.

The vertex-sector curvature is the standard graph Laplacian $A_n^{\text{vertex}} := d_n^* d_n$ on X_n^0 , with $d_n : X_n^0 \rightarrow X_n^1$ the oriented difference (with respect to an arbitrary but fixed orientation of E_n).

In this abstract model, the coboundary refinement relation

$$p_{n+1,n}^1 d_{n+1} = \frac{1}{2} d_n p_{n+1,n}^0 \quad (1)$$

holds verbatim, by direct computation: for any $f \in X_{n+1}^0$, $(p^1 d_{n+1} f)(e) = \frac{1}{2}[(d_{n+1} f)(u, m_e) + (d_{n+1} f)(m_e, v)] = \frac{1}{2}[(f(m_e) - f(u)) + (f(v) - f(m_e))] = \frac{1}{2}(f(v) - f(u)) = \frac{1}{2}(d_n p^0 f)(e)$. We use (1) as the load-bearing input below. The relation also holds in the source's full Schläfli setting (P4 **coboundary-refinement**, lines 896–937), but the present paper only uses the abstract-model version, where it is elementary.

The gap to the full P3/P4 tower. The Theorems and Proposition below all hold only *within Definition 1.1's abstract model*. They do *not* immediately transfer to the full Schläfli setting of [3]. The minimal lift hypotheses we identify are:

- (L1) the level- $(n+1)$ vertex Laplacian, restricted to boundary vertices $V_n \subset V(G_{n+1}^\bullet)$, agrees with the Laplacian of the abstract model up to a defect that is supported only on midpoints (no extra contribution from centroid neighbours or centroid-midpoint edges);
- (L2) the parent-fibre averaging map $p_{n+1,n}^1$ on F^1 -valued, antisymmetric oriented-edge chains satisfies the calibration identity $p^1(p^1)^* = \frac{1}{2} I$ in the F^1 inner product, with the same factor of $\frac{1}{2}$ as in the scalar abstract model;
- (L3) the harmonic-extension operator H_{n+1} in the F^0 -valued setting saturates the energy lower bound coming from (L1) and (L2).

Each of these is concrete and isolatable: (L1) is a graph-theoretic incidence statement about the $W^\bullet$ -equivariant midpoint and centroid placement; (L2) is an inner-product calculation in the source's F^1 -valued, antisymmetric oriented-edge convention (**cascade-refinement-spaces.tex** lines 464–493); (L3) is the analogue of Step 3 of the proof of Theorem 2.1 in the F -valued setting. Lifting the Schur-halving identity to the full P3/P4 tower under these hypotheses is named here as the remaining upstream task and is not attempted in this paper.

1.2 Source objects (recap)

Within the abstract model of Definition 1.1, we use scalar finite-dimensional chain spaces. The vertex-block curvature operator is

$$A_n^{\text{vertex}} := d_n^* d_n \in \mathbb{R}^{V_n \times V_n}, \quad (2)$$

the standard graph Laplacian on V_n .

The bonding maps are the pair $p_{n+1,n} = (p_{n+1,n}^0, p_{n+1,n}^1)$, where $p_{n+1,n}^0 : X_{n+1}^0 \rightarrow X_n^0$ is vertex-restriction along the canonical inclusion $V_n \subset V_{n+1}$ and $p_{n+1,n}^1 : X_{n+1}^1 \rightarrow X_n^1$ is the parent-fibre average by factor $\frac{1}{2}$ on each of the two children of every parent edge.

The coboundary refinement relation (1) of §1.1 is the load-bearing input below. (The analogous identity is also a theorem in the source's full Schläfli setting, [2] Proposition **coboundary-refinement** at lines 896–937.)

1.3 What this paper proves

The paper's central object is the scaled vertex curvature operator:

Definition 1.2 (Scaled curvature operator). $\tilde{A}_n := 2^n A_n^{\text{vertex}}$, defined on X_n^0 . The corresponding scaled residual is $\tilde{R}_n(\phi) := \frac{1}{2} \langle \phi, \tilde{A}_n \phi \rangle = 2^{n-1} \langle \phi, A_n^{\text{vertex}} \phi \rangle$.

The scaling factor 2^n is the unique factor that compensates the refinement-induced halving in (1) once one passes from the coboundary level to the curvature (d^*d) level. The main results are an exact Schur-complement identity for $\tilde{A}_n$ under boundary–interior decomposition (Theorem 2.1), a clean energy-intertwining inequality on the harmonic-extension sector that discharges cascade-mechanism's (O3) (Theorem 3.1), and an explicit defect identity showing that strict operator-level refinement-compatibility for $\tilde{A}_n$ remains false (Proposition 4.1).

2 Schur-complement identity

Fix a level n and a finite refinement $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$ such that $|V(G_{n+1}^\bullet)| - |V(G_n^\bullet)|$ new “midpoint” vertices are added (each created as the midpoint of one existing edge of $G_n^\bullet$ that was subdivided to produce $G_{n+1}^\bullet$). Partition the vertex set $V(G_{n+1}^\bullet) = B \sqcup I$ into

- $B := V(G_n^\bullet)$, the *boundary* vertices (those retained from level n);
- $I := V(G_{n+1}^\bullet) \setminus V(G_n^\bullet)$, the *interior* or *new* vertices.

Block-decompose A_{n+1}^{vertex} in this (B, I) ordering:

$$A_{n+1}^{\text{vertex}} = \begin{pmatrix} A_{BB} & A_{BI} \\ A_{IB} & A_{II} \end{pmatrix}, \quad (3)$$

where $A_{BB} \in \mathbb{R}^{B \times B}$, $A_{BI} = A_{IB}^T \in \mathbb{R}^{B \times I}$, $A_{II} \in \mathbb{R}^{I \times I}$.

By construction A_n^{vertex} is the graph Laplacian on $V(G_n^\bullet)$. The harmonic extension $H_{n+1} : X_n^0 \rightarrow X_{n+1}^0$ is the unique linear map sending $f \in X_n^0$ to the level- $(n+1)$ state with prescribed boundary values f and interior values $f_I = -A_{II}^{-1} A_{IB} f$, equivalently the unique minimiser of the level- $(n+1)$ energy $\frac{1}{2} \langle \cdot, A_{n+1}^{\text{vertex}} \cdot \rangle$ subject to the boundary constraint $p_{n+1,n}^0 = f$. We assume A_{II} is invertible (the standard finite-dimensional case in [3, 2]).

The Schur complement of A_{n+1}^{vertex} with respect to the partition (B, I) is

$$\text{Schur}(A_{n+1}^{\text{vertex}}) := A_{BB} - A_{BI} A_{II}^{-1} A_{IB} \in \mathbb{R}^{B \times B}. \quad (4)$$

Theorem 2.1 (Schur-complement halving in the abstract model). *In the abstract scalar pure-midpoint refinement model of Definition 1.1, with vertex-sector curvature $A_n^{\text{vertex}} = d_n^* d_n$, the Schur complement of A_{n+1}^{vertex} under the (boundary, interior) partition (3) is exactly half the level- n Laplacian:*

$$\text{Schur}(A_{n+1}^{\text{vertex}}) = \frac{1}{2} A_n^{\text{vertex}}. \quad (5)$$

Equivalently, with $\tilde{A}_n := 2^n A_n^{\text{vertex}}$:

$$\text{Schur}(\tilde{A}_{n+1}) = \tilde{A}_n. \quad (6)$$

Proof. We proceed in three steps: identify the Schur complement with a harmonic energy, decompose that energy via the source's coboundary refinement relation, and verify directly that the harmonic extension saturates the resulting lower bound.

Step 1. For any positive-semi-definite A_{n+1}^{vertex} with invertible interior block A_{II} , the Schur complement and the harmonic-extension energy agree: $\langle f, \text{Schur}(A_{n+1}^{\text{vertex}}) f \rangle = \min_{\psi: p^0 \psi = f} \langle \psi, A_{n+1}^{\text{vertex}} \psi \rangle =$

$\langle H_{n+1}f, A_{n+1}^{\text{vertex}} H_{n+1}f \rangle$ for every $f \in X_n^0$. This is standard: for $\psi = (f, f_I)$ in block form, $\langle \psi, A_{n+1}^{\text{vertex}} \psi \rangle = \langle f, A_{BB}f \rangle + 2\langle f_I, A_{IB}f \rangle + \langle f_I, A_{II}f_I \rangle$ is minimised in f_I at $f_I^* = -A_{II}^{-1}A_{IB}f$ (since A_{II} is positive definite), with minimum value $\langle f, (A_{BB} - A_{BI}A_{II}^{-1}A_{IB})f \rangle$, and $H_{n+1}f := (f, f_I^*)$ by definition of harmonic extension.

Step 2. Let $f \in X_n^0$ and let $\psi \in X_{n+1}^0$ satisfy $p^0\psi = f$. Apply the coboundary refinement relation (1): $p^1 d_{n+1}\psi = \frac{1}{2}d_n p^0\psi = \frac{1}{2}d_n f$, which is fixed independently of ψ (subject to $p^0\psi = f$). The averaging-by- $\frac{1}{2}$ structure of p^1 on a 2-element fibre yields the calibration identity

$$p^1(p^1)^* = \frac{1}{2} I_{X_n^1}, \quad (7)$$

which is verified directly: for any $\omega \in X_n^1$ and any level- n edge e with two children e_1, e_2 , one has $((p^1)^*\omega)(e_i) = \frac{1}{2}\omega(e)$ (the unique adjoint under standard ℓ^2 inner products) and $(p^1(p^1)^*\omega)(e) = \frac{1}{2} \cdot \frac{1}{2}\omega(e) + \frac{1}{2} \cdot \frac{1}{2}\omega(e) = \frac{1}{2}\omega(e)$. Hence the orthogonal projection of any $\xi \in X_{n+1}^1$ onto $\text{range}((p^1)^*)$ is $P\xi = (p^1)^*((p^1(p^1)^*)^{-1}p^1\xi) = 2(p^1)^*p^1\xi$, and X_{n+1}^1 decomposes orthogonally as $\text{range}((p^1)^*) \oplus \ker p^1$. Apply this decomposition to $\xi = d_{n+1}\psi$:

$$d_{n+1}\psi = (p^1)^*d_n f + r_\psi, \quad r_\psi \in \ker p^1, \quad (8)$$

where the first term comes from $P(d_{n+1}\psi) = 2(p^1)^*(p^1 d_{n+1}\psi) = 2(p^1)^*(\frac{1}{2}d_n f) = (p^1)^*d_n f$. Compute the norm using (7):

$$\|(p^1)^*d_n f\|^2 = \langle (p^1)^*d_n f, (p^1)^*d_n f \rangle = \langle d_n f, p^1(p^1)^*d_n f \rangle = \frac{1}{2}\|d_n f\|^2 = \frac{1}{2}\langle f, A_n^{\text{vertex}} f \rangle.$$

By orthogonality of the decomposition (8),

$$\langle \psi, A_{n+1}^{\text{vertex}} \psi \rangle = \|d_{n+1}\psi\|^2 = \frac{1}{2}\langle f, A_n^{\text{vertex}} f \rangle + \|r_\psi\|^2 \geq \frac{1}{2}\langle f, A_n^{\text{vertex}} f \rangle. \quad (9)$$

Step 3. We exhibit one specific ψ with $r_\psi = 0$, hence equality in (9): the harmonic extension itself. By the edge-midpoint hypothesis of the theorem statement, every interior vertex $m_e \in I$ has degree exactly 2 in $G_{n+1}^\bullet$, its only neighbours being the two endpoints u, v of its parent edge $e \in E(G_n^\bullet)$; hence the harmonic-extension formula $f_I = -A_{II}^{-1}A_{IB}f$ at m_e reduces to the elementary two-neighbour average $(H_{n+1}f)(m_e) = \frac{1}{2}(f(u) + f(v))$. Compute $d_{n+1}(H_{n+1}f)$ at the two child edges of e :

$$\begin{aligned} (d_{n+1}(H_{n+1}f))(u, m_e) &= (H_{n+1}f)(m_e) - (H_{n+1}f)(u) = \frac{1}{2}(f(u) + f(v)) - f(u) = \frac{1}{2}(f(v) - f(u)) = \frac{1}{2}(d_n f)(e), \\ (d_{n+1}(H_{n+1}f))(m_e, v) &= (H_{n+1}f)(v) - (H_{n+1}f)(m_e) = f(v) - \frac{1}{2}(f(u) + f(v)) = \frac{1}{2}(f(v) - f(u)) = \frac{1}{2}(d_n f)(e). \end{aligned}$$

On the other hand, $((p^1)^*d_n f)(u, m_e) = \frac{1}{2}(d_n f)(e) = ((p^1)^*d_n f)(m_e, v)$ by definition of $(p^1)^*$ on a two-element fibre. Hence $d_{n+1}(H_{n+1}f) = (p^1)^*d_n f$ on every level- $(n+1)$ edge, i.e., $r_{H_{n+1}f} = 0$.

Combining Steps 1–3:

$$\langle f, \text{Schur}(A_{n+1}^{\text{vertex}})f \rangle = \langle H_{n+1}f, A_{n+1}^{\text{vertex}} H_{n+1}f \rangle = \frac{1}{2}\langle f, A_n^{\text{vertex}} f \rangle + \|r_{H_{n+1}f}\|^2 = \frac{1}{2}\langle f, A_n^{\text{vertex}} f \rangle.$$

This holds for every $f \in X_n^0$. Since $\text{Schur}(A_{n+1}^{\text{vertex}})$ and A_n^{vertex} are both symmetric, polarisation of the quadratic form gives the operator identity (5). The scaled form (6) follows by linearity: Schur is a quadratic operation but its dependence on the overall scale of the input matrix is linear in that scale, so $\text{Schur}(\tilde{A}_{n+1}) = \text{Schur}(2^{n+1}A_{n+1}^{\text{vertex}}) = 2^{n+1}\text{Schur}(A_{n+1}^{\text{vertex}}) = 2^{n+1} \cdot \frac{1}{2}A_n^{\text{vertex}} = 2^n A_n^{\text{vertex}} = \tilde{A}_n$. $\square$

Remark 2.2 (Numerical verification). Theorem 2.1 is verified to exact rational precision in §6 for two finite refinement instances: the single edge with one-midpoint subdivision (Tower I, $|V(G_0^\bullet)| = 2$, $|V(G_1^\bullet)| = 3$) and the triangle with edge-midpoint subdivision (Tower II, $|V(G_0^\bullet)| = 3$, $|V(G_1^\bullet)| = 6$). The numerical witness is `papers/cascade-refinement-compat/scripts/explore_defect.py`; all linear algebra is exact `Fraction` arithmetic, with no floating-point error.

3 Energy intertwining and discharge of (O3)

We now apply Theorem 2.1 to the cascade-mechanism consumer-API obligations. The cascade-mechanism residual functional in this paper's setting is $\tilde{R}_n(\phi) := \frac{1}{2}\langle\phi, \tilde{A}_n\phi\rangle$.

Theorem 3.1 (Energy intertwining; discharge of (O3)). *In the abstract scalar pure-midpoint refinement model of Definition 1.1, for every $\psi \in X_{n+1}^0$,*

$$\tilde{R}_n(p_{n+1,n}^0\psi) \leq \tilde{R}_{n+1}(\psi), \quad (10)$$

with equality if and only if $\psi = H_{n+1}(p_{n+1,n}^0\psi)$ (i.e. ψ is the harmonic extension of its boundary).

Proof. Decompose $\psi = H_{n+1}(p_{n+1,n}^0\psi) + \psi^\perp$, where $\psi^\perp := \psi - H_{n+1}(p_{n+1,n}^0\psi)$ vanishes on $B = V(G_n^\bullet)$ (since $p_{n+1,n}^0 H_{n+1}(p_{n+1,n}^0\psi) = p_{n+1,n}^0\psi$). The energy quadratic form decomposes as

$$\begin{aligned} \langle\psi, A_{n+1}^{\text{vertex}}\psi\rangle &= \langle H_{n+1}(p^0\psi), A_{n+1}^{\text{vertex}} H_{n+1}(p^0\psi)\rangle \\ &\quad + 2\langle H_{n+1}(p^0\psi), A_{n+1}^{\text{vertex}}\psi^\perp\rangle + \langle\psi^\perp, A_{n+1}^{\text{vertex}}\psi^\perp\rangle. \end{aligned}$$

The cross term vanishes: writing A_{n+1}^{vertex} in block form (3) and noting that $\psi^\perp$ has zero B -component, one has $A_{n+1}^{\text{vertex}}\psi^\perp = (A_{BI}\psi_I^\perp, A_{II}\psi_I^\perp)$, and $H_{n+1}(p^0\psi) = (p^0\psi, -A_{II}^{-1}A_{IB}p^0\psi)$; the cross-pairing computes to $\langle p^0\psi, A_{BI}\psi_I^\perp\rangle + \langle -A_{II}^{-1}A_{IB}p^0\psi, A_{II}\psi_I^\perp\rangle = \langle p^0\psi, A_{BI}\psi_I^\perp\rangle - \langle p^0\psi, A_{BI}\psi_I^\perp\rangle = 0$. The first surviving term equals $\langle p^0\psi, \text{Schur}(A_{n+1}^{\text{vertex}})p^0\psi\rangle = \frac{1}{2}\langle p^0\psi, A_n^{\text{vertex}}p^0\psi\rangle$ by Theorem 2.1. The third surviving term is $\langle\psi_I^\perp, A_{II}\psi_I^\perp\rangle \geq 0$ since A_{II} is positive semi-definite (a principal submatrix of the positive-semi-definite A_{n+1}^{vertex}). Hence

$$\langle\psi, A_{n+1}^{\text{vertex}}\psi\rangle \geq \frac{1}{2}\langle p^0\psi, A_n^{\text{vertex}}p^0\psi\rangle.$$

Multiplying both sides by 2^n : $2^n\langle\psi, A_{n+1}^{\text{vertex}}\psi\rangle \geq 2^{n-1}\langle p^0\psi, A_n^{\text{vertex}}p^0\psi\rangle$, which after dividing by 2 and re-grouping factors becomes $\langle\psi, \tilde{A}_{n+1}\psi\rangle/2 \geq \langle p^0\psi, \tilde{A}_n p^0\psi\rangle/2$, i.e. $\tilde{R}_{n+1}(\psi) \geq \tilde{R}_n(p^0\psi)$, which is (10). Equality requires $\psi^\perp = 0$ (since A_{II} is positive definite by invertibility), i.e. ψ is the harmonic extension of its boundary. $\square$

Corollary 3.2 ((O3) is closed in the abstract pure-midpoint model). *In the abstract scalar pure-midpoint refinement model of Definition 1.1, the cascade-mechanism residual-monotonicity hypothesis [1] (O3) holds for the scaled curvature residual $\tilde{R}_n(\phi) = \frac{1}{2}\langle\phi, \tilde{A}_n\phi\rangle$, where $\tilde{A}_n = 2^n A_n^{\text{vertex}}$ and $A_n^{\text{vertex}} = d_n^* d_n$ is the vertex graph Laplacian. Lifting this discharge to the full Schläfli P3/P4 tower of [2] (which adds face-centroid vertices and centroid-midpoint edges absent from Definition 1.1) is open and named in §1.1.*

4 Strict operator-level refinement-compatibility fails

Theorem 2.1 closes the energy-level refinement question. The strict operator-level identity $p_{n+1,n}^0 \tilde{A}_{n+1} = \tilde{A}_n p_{n+1,n}^0$ remains false, with explicit defect.

Proposition 4.1 (Strict operator intertwining fails; kernel of the defect). *In the abstract scalar pure-midpoint refinement model of Definition 1.1, the strict identity $p_{n+1,n}^0 \tilde{A}_{n+1} = \tilde{A}_n p_{n+1,n}^0$ does not hold in general, and equivalently $p_{n+1,n}^0 A_{n+1}^{\text{vertex}} \neq \frac{1}{2} A_n^{\text{vertex}} p_{n+1,n}^0$ as operators $X_{n+1}^0 \rightarrow X_n^0$. The defect operator*

$$D_n := p_{n+1,n}^0 A_{n+1}^{\text{vertex}} - \frac{1}{2} A_n^{\text{vertex}} p_{n+1,n}^0 \quad (11)$$

satisfies the following:

- (i) $D_n \psi = 0$ whenever $\psi \in \text{range}(H_{n+1})$, i.e. whenever ψ is the harmonic extension of its own boundary $p^0 \psi$;
- (ii) Writing $\delta(\psi) \in \mathbb{R}^{E_n}$ for the per-edge midpoint deficit $\delta_e(\psi) := \frac{1}{2}(\psi_u + \psi_v) - \psi_{m_e}$ (with $e = \{u, v\} \in E_n$ and m_e the midpoint inserted on e), one has $(D_n \psi)_v = \sum_{e \ni v} \delta_e(\psi)$ for every $v \in V_n$. Equivalently, $D_n \psi = B_n \delta(\psi)$ where $B_n \in \{0, 1\}^{V_n \times E_n}$ is the unsigned vertex-edge incidence matrix of G_n . Consequently

$$\ker D_n = \text{range}(H_{n+1}) + \{\psi : \delta(\psi) \in \ker B_n, \psi_B = 0\}, \quad (12)$$

with the two summands intersecting trivially. The kernel $\ker B_n$ is non-trivial precisely when G_n admits an alternating-sign edge-weighting along a closed walk of even length; in particular $\ker B_n = 0$ when G_n is a forest, and $\ker D_n = \text{range}(H_{n+1})$ in that case. The smallest counterexample is $G_n = C_4$ (4-cycle), where $\dim \ker B_n = 1$ and $\ker D_n$ is strictly larger than $\text{range}(H_{n+1})$;

- (iii) D_n does not vanish on the level- n -supported subspace $\{\psi : \psi_I = 0\}$ in general (and not at all when G_n has at least one edge).

Proof. Per-vertex formula. Fix $v \in V_n$. By Definition 1.1 (M2), the level- $(n+1)$ neighbours of v are exactly the midpoints $\{m_e : e \in E_n, e \ni v\}$ of the level- n edges incident to v , and v has the same level- $(n+1)$ degree $\deg_{G_{n+1}}(v) = \deg_{G_n}(v)$. Hence

$$(A_{n+1}^{\text{vertex}} \psi)(v) = \sum_{e \ni v} (\psi(v) - \psi(m_e)) \quad \text{and} \quad (A_n^{\text{vertex}} \psi_B)(v) = \sum_{e=\{v,u\} \ni v} (\psi(v) - \psi(u)),$$

so for every $\psi \in X_{n+1}^0$ with $\psi_B = p^0 \psi$,

$$\begin{aligned} (D_n \psi)(v) &= (A_{n+1}^{\text{vertex}} \psi)(v) - \frac{1}{2} (A_n^{\text{vertex}} \psi_B)(v) \\ &= \sum_{e \ni v} (\psi(v) - \psi(m_e)) - \frac{1}{2} \sum_{e=\{v,u\}} (\psi(v) - \psi(u)) \\ &= \sum_{e \ni v} \left[\frac{1}{2} (\psi(u) + \psi(v)) - \psi(m_e) \right] = \sum_{e \ni v} \delta_e(\psi) = (B_n \delta(\psi))(v). \end{aligned}$$

This proves the formula in the proposition statement.

(i) If $\psi = H_{n+1} f$ is the harmonic extension of $f \in X_n^0$, the harmonic-equation $(A_{II})_{m_e, m_e} \psi(m_e) + (A_{IB})_{m_e, \cdot} f = 0$ collapses, since m_e has level- $(n+1)$ neighbours $\{u, v\}$ (so A_{II} is diagonal with entries 2 on midpoints, A_{IB} has rows $-e_u^T - e_v^T$ on midpoint m_e), to $2\psi(m_e) = f(u) + f(v)$, i.e. $\psi(m_e) = \frac{1}{2}(f(u) + f(v))$. Hence $\delta_e(\psi) = \frac{1}{2}(f(u) + f(v)) - \psi(m_e) = 0$ for all $e \in E_n$, and the per-vertex formula gives $D_n \psi = B_n \cdot 0 = 0$.

(ii) From the per-vertex formula above, $D_n \psi = 0$ iff $B_n \delta(\psi) = 0$ iff $\delta(\psi) \in \ker B_n$. The map $\psi \mapsto \delta(\psi)$ is surjective $X_{n+1}^0 \twoheadrightarrow \mathbb{R}^{E_n}$ (any prescribed $\delta \in \mathbb{R}^{E_n}$ is realised by $\psi_B := 0$ and $\psi_{m_e} := -\delta_e$), and its kernel is exactly $\text{range}(H_{n+1})$ by the harmonic-extension equation $\psi(m_e) = \frac{1}{2}(\psi(u) + \psi(v))$ in part (i). Hence the short exact sequence $0 \rightarrow \text{range}(H_{n+1}) \rightarrow \ker D_n \rightarrow \ker B_n \rightarrow 0$ splits via the section $\delta \mapsto \psi$ defined by $\psi_B = 0$, $\psi_{m_e} = -\delta_e$, giving (12). The non-triviality of $\ker B_n$ on closed even-length walks is the standard fact that the unsigned incidence matrix has $\ker B_n^T B_n = \ker B_n$ with $B_n^T B_n = 2I_{E_n} + A_{\text{line}}^{\text{adj}}(G_n)$ on the line graph, whose null space is generated by alternating-sign ± 1 patterns along even closed walks (and is trivial on trees / forests). For $G_n = C_4$, an explicit kernel generator is $\delta = (1, -1, 1, -1)$ on the four edges in cyclic order.

(iii) For any $\phi \in X_{n+1}^0$ with $\phi_I = 0$ and edge $e = \{u, v\} \in E_n$, $\delta_e(\phi) = \frac{1}{2}(\phi(u) + \phi(v))$. Take any ϕ with $\phi(u) \neq -\phi(v)$ on at least one edge (possible whenever G_n has an edge): then $\delta(\phi) \neq 0$,

and the per-vertex formula gives $D_n\phi = B_n\delta(\phi)$ with the right-hand side non-zero at any vertex incident to an edge e where $\delta_e(\phi) \neq 0$ and not cancelled by sums at that vertex. The explicit defect $D_n\phi$ is computed for two concrete towers in §6; in both, $D_n|_{\{\phi_I=0\}}$ is non-zero. $\square$

Remark 4.2 (Geometric interpretation). The conclusion (i) of Proposition 4.1 is the operator-level shadow of Theorem 2.1: harmonic extensions saturate the boundary Laplacian action under p^0 at the strict scaled level. Conclusion (ii) records the genuinely finer fact that, on parent graphs G_n with non-trivial unsigned incidence kernel (the smallest example is $G_n = C_4$), the kernel of D_n is strictly larger than $\text{range}(H_{n+1})$: there exist non-harmonic level- $(n+1)$ states whose midpoint deficits form a non-zero edge-cocycle in $\ker B_n$ but whose total at every boundary vertex still cancels. The defect D_n measures the boundary-vertex sum of midpoint deficits $\delta_e(\psi) = \frac{1}{2}(\psi_u + \psi_v) - \psi_{m_e}$ along incident edges, not the per-edge midpoint deficit itself.

Remark 4.3 (Energy vs. flow level). The Schur identity Theorem 2.1 and its energy consequence Theorem 3.1 hold at the *quadratic-form / boundary-energy* level. Strict *operator-level* intertwining $p^0\tilde{A}_{n+1} = \tilde{A}_n p^0$ fails by Proposition 4.1, and consequently the strict *flow-level* intertwining $p^0 \circ e^{-t\tilde{A}_{n+1}} = e^{-t\tilde{A}_n} \circ p^0$ also fails. Cascade-mechanism’s obligation (O2), as stated, is at the flow level. The natural source-supported reformulation of (O2) is the Schur-energy version, which Theorem 2.1 discharges. In the cascade-mechanism reformulation table of §7 we record this as a *partial* discharge of (O2).

5 Kernel of A_n^{vertex} and (O1)

For completeness we record the status of cascade-mechanism’s obligation (O1) (top-rung residual-zero fixed point and convergence) in the abstract scalar pure-midpoint refinement model of Definition 1.1, applied at the top level $n = N$.

Proposition 5.1 (Kernel and convergence of $\tilde{A}_N$). *For any finite level N , $\tilde{A}_N$ is positive semi-definite on X_N^0 and its kernel $\ker \tilde{A}_N = \ker A_N^{\text{vertex}}$ consists exactly of the locally-constant functions on the connected components of $G_N^\bullet$. If $G_N^\bullet$ is connected, $\dim \ker \tilde{A}_N = 1$ (constants only).*

For every initial datum $\phi^{(0)} \in X_N^0$, the gradient flow $\Psi_N^t := e^{-t\tilde{A}_N}$ converges as $t \rightarrow \infty$ to the orthogonal projection of $\phi^{(0)}$ onto $\ker \tilde{A}_N$. This is automatic in finite dimensions by spectral decomposition.

In particular, the trivial state $\phi_N^ = 0$ is always a residual-zero fixed point, and the cascade-mechanism obligation (O1) “nontrivial-selected-state existence” reduces to whether $\ker \tilde{A}_N \neq \{0\}$ (yes, in the connected case the kernel is one-dimensional) and whether the projection of the initial datum onto the kernel is non-zero (which is generic but not automatic).*

Proof. $A_N^{\text{vertex}} = d_N^* d_N$ is positive semi-definite by construction, with kernel $\ker d_N$, the locally-constant functions on the connected components of $G_N^\bullet$. The convergence statement for $e^{-tA_N^{\text{vertex}}}$ is the standard finite-dimensional functional-calculus result for a self-adjoint positive-semi-definite operator. Multiplying by the positive scalar 2^N does not change the kernel or the convergence behaviour. $\square$

Remark 5.2 (Selection rule is not part of cascade closure). [1] §5 explicitly states that selection of the top-rung fixed point is not provided by cascade closure and is the successor paper’s own task. The kernel description above records what cascade closure inherits at the top rung; the selection rule by which a particular non-zero element of $\ker \tilde{A}_N$ is preferred over the zero state is a question for downstream papers, and outside the scope of the present source-side discharge.

6 Worked finite refinement towers

This section gives explicit matrix-level computations on two concrete finite refinement towers, verifying Theorems 2.1 and 3.1 and Proposition 4.1 to exact rational precision. The numerical witness is the script `papers/cascade-refinement-compat/scripts/explore_defect.py`; its output is preserved in `papers/cascade-refinement-compat/repro/expected_outputs/`.

6.1 Tower I: single edge $\rightarrow$ one-midpoint subdivision

Let $G_0^\bullet$ be the graph on $V(G_0^\bullet) = \{u, v\}$ with the single edge $e = (u, v)$. Refine by inserting the midpoint m of e , splitting e into two new edges (u, m) and (m, v) : $V(G_1^\bullet) = \{u, m, v\}$, $E(G_1^\bullet) = \{(u, m), (m, v)\}$. The discrete coboundaries in oriented form are $d_0 = \begin{pmatrix} -1 & 1 \end{pmatrix}$ on $X_0^0 = \mathbb{R}^2$, and $d_1 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$ on $X_1^0 = \mathbb{R}^3$. The bonding maps are $p_{1,0}^0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $p_{1,0}^1 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \end{pmatrix}$. The vertex Laplacians are

$$A_0^{\text{vertex}} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad A_1^{\text{vertex}} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

With the partition $B = \{u, v\}$, $I = \{m\}$, the block decomposition yields $A_{BB} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $A_{BI} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$, $A_{IB} = \begin{pmatrix} -1 & -1 \end{pmatrix}$, $A_{II} = (2)$. The Schur complement is

$$\text{Schur}(A_1^{\text{vertex}}) = A_{BB} - A_{BI}A_{II}^{-1}A_{IB} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} = \frac{1}{2}A_0^{\text{vertex}}.$$

This is exactly Theorem 2.1.

The defect of strict refinement-compatibility is

$$D_0 := p_{1,0}^0 A_1^{\text{vertex}} - \frac{1}{2} A_0^{\text{vertex}} p_{1,0}^0 = \begin{pmatrix} 1 & -1 & 0 \\ 0 & -1 & 1 \end{pmatrix} - \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -1 & \frac{1}{2} \\ \frac{1}{2} & -1 & \frac{1}{2} \end{pmatrix}.$$

Both rows of D_0 act on $\phi = (\phi(u), \phi(m), \phi(v))$ as $\frac{1}{2}\phi(u) - \phi(m) + \frac{1}{2}\phi(v) = \frac{1}{2}(\phi(u) + \phi(v)) - \phi(m)$, the “midpoint deficit” of ϕ at m . Tower I has a single edge with no cycles, so the unsigned vertex-edge incidence matrix $B_0 \in \{0, 1\}^{2 \times 1}$ has trivial kernel, and the incidence characterisation in Proposition 4.1(ii) collapses to: $D_0\phi = 0$ iff the midpoint deficit at m is zero, iff $\phi(m) = \frac{1}{2}(\phi(u) + \phi(v))$, iff ϕ is the harmonic extension of its boundary. So here the kernel of D_0 coincides with $\text{range}(H_1)$. Furthermore, D_0 does *not* vanish on $\{\phi : \phi(m) = 0\}$ (which gives $D_0\phi = (\frac{1}{2}(\phi(u) + \phi(v)), \frac{1}{2}(\phi(u) + \phi(v)))$, generically non-zero), confirming Proposition 4.1(iii).

6.2 Tower II: triangle $\rightarrow$ edge-midpoint subdivision

Let $G_0^\bullet$ be the 3-cycle on $V(G_0^\bullet) = \{0, 1, 2\}$ with edges $e_{01} = (0, 1)$, $e_{12} = (1, 2)$, $e_{20} = (2, 0)$. Refine by inserting one midpoint per edge, m_{01}, m_{12}, m_{20} , giving $V(G_1^\bullet) = \{0, 1, 2, m_{01}, m_{12}, m_{20}\}$ ($|V(G_1^\bullet)| = 6$) and the six new edges $(0, m_{01}), (m_{01}, 1), (1, m_{12}), (m_{12}, 2), (2, m_{20}), (m_{20}, 0)$. With $B = \{0, 1, 2\}$, $I = \{m_{01}, m_{12}, m_{20}\}$ and the identification of bonding maps $p_{1,0}^0, p_{1,0}^1$ analogously to Tower I, direct computation (verified by `explore_defect.py`) yields

$$A_0^{\text{vertex}} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

and the Schur complement

$$\text{Schur}(A_1^{\text{vertex}}) = \begin{pmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & 1 \end{pmatrix} = \frac{1}{2}A_0^{\text{vertex}},$$

again confirming Theorem 2.1. Eigenvalues of the defect operator $\text{Schur}(A_1^{\text{vertex}}) - A_0^{\text{vertex}}$ on this tower are $\{0, -\frac{3}{2}, -\frac{3}{2}\}$, confirming the ordering $\text{Schur} \leq A_0^{\text{vertex}}$ (i.e. the unscaled Schur complement under-estimates the level-0 Laplacian; rescaling by 2^n at level n exactly compensates).

7 Cascade-mechanism update

The present paper updates the worked-invocation status table of [1] §3.5 in the abstract scalar pure-midpoint refinement model of Definition 1.1, which §1.1 relates to the imported P3/P4 finite-level refinement tower of [2]: any update lifts to that tower under the named hypotheses (L1)–(L3) of §1.1 and otherwise stays inside the abstract model. The consumer-API obligations themselves ([1] §3.4) are unchanged; the cascade-closure mechanism, Definition, Proposition, and Corollary of [1] are unchanged.

Table 1: Updated status of cascade-mechanism consumer-API obligations in the abstract scalar pure-midpoint refinement model (Definition 1.1) with the scaled curvature operator $\tilde{A}_n = 2^n A_n^{\text{vertex}}$. Lifting any row to the full P3/P4 tower of [2] requires the named hypotheses (L1)–(L3) of §1.1.

Obligation	Status (this paper)	Where established
(O0) Standing analytic / admissibility hypotheses	Closed. Finite-dimensional Hilbert structure of X_n^0 , C^1 bonding maps, well-posed flow $e^{-t\tilde{A}_n}$.	[3]; recap in §1.
(O1) Top-rung fixed point and convergence	Closed at the trivial level (zero state always residual-zero); nontrivial selection reduces to: kernel of $\tilde{A}_N$ is the constants on connected components of $G_N^\bullet$, and convergence is automatic by spectral decomposition. The selection rule itself is out of scope per [1] §5.	Proposition 5.1.
(O2) Flow intertwining	Closed at the Schur / quadratic-form level only. Strict operator-level identity $p^0 \tilde{A}_{n+1} = \tilde{A}_n p^0$ remains false (Proposition 4.1); strict flow-level identity also remains false. The boundary-energy form satisfies $\text{Schur}(\tilde{A}_{n+1}) = \tilde{A}_n$ exactly (Theorem 2.1).	Theorem 2.1, Proposition 4.1.

Obligation	Status (this paper)	Where established
(O3) Residual monotonicity	Closed. $\tilde{R}_n(p^0\psi) \leq \tilde{R}_{n+1}(\psi)$ for every $\psi \in X_{n+1}^0$, with equality iff ψ is the harmonic extension of its boundary.	Theorem 3.1, Corollary 3.2.

The discharge of (O3) and the partial discharge of (O2) are non-trivial relative to [1]’s original status report; the structural discovery is that the source’s coboundary factor- $\frac{1}{2}$ relation (1) lifts to the curvature level via the Schur complement, with the scaled operator $\tilde{A}_n := 2^n A_n^{\text{vertex}}$ as the natural refinement-compatible object at the boundary-energy level.

8 Discussion

What this paper does. It proves that, in the abstract scalar pure-midpoint refinement model of Definition 1.1 and with the scaled curvature operator $\tilde{A}_n = 2^n A_n^{\text{vertex}}$, the cascade-mechanism residual-monotonicity obligation (O3) holds (Theorem 3.1, Corollary 3.2) and the flow-intertwining obligation (O2) holds at the boundary-energy / Schur level (Theorem 2.1). The mechanism by which both happen is the same: the coboundary factor- $\frac{1}{2}$ relation $p^1 d_{n+1} = \frac{1}{2} d_n p^0$, which holds verbatim in the abstract model (and as a theorem in the source’s full Schläfli setting), pulled forward through the boundary–interior decomposition.

What this paper does not do. It does not establish strict operator-level $p^0 \tilde{A}_{n+1} = \tilde{A}_n p^0$ intertwining; that identity is genuinely false in the abstract model. The defect operator D_n defined in Proposition 4.1 satisfies $\text{range}(H_{n+1}) \subseteq \ker D_n$, with equality when the parent graph G_n has no bipartite component, and D_n is non-zero on the level- n -supported sector $\{\psi : \psi_I = 0\}$ in general. In particular, the strict cascade-mechanism flow-intertwining hypothesis (O2) at the operator/flow level remains a strictly stronger condition than what the abstract model supports; cascade-mechanism users invoking the boundary-energy version of (O2) will find it discharged here, but those requiring strict operator-level intertwining will find the defect computed but not zero. The paper also does not lift any of these results to the full Schläfli P3/P4 tower of [2]: that lift is named in §1.1 as hypotheses (L1)–(L3) and is not attempted here. The paper does not derive a selection rule for nontrivial elements of $\ker \tilde{A}_N$, in keeping with [1] §5’s explicit boundary.

Implication for cascade-mechanism v2.1. The minimum cascade-mechanism update is a single sentence in [1] §3.5 replacing the “(O3) open” line with a citation of Corollary 3.2 above, scoped to the abstract pure-midpoint model of Definition 1.1, and a short remark recording the partial discharge of (O2) at the Schur level under the same scope. The cascade-mechanism consumer API itself ([1] §3.4 (O0)–(O3)) does not change. The reproduction bundle of [1] is unaffected. Any update describing the full P3/P4 tower must be flagged conditional on (L1)–(L3) of §1.1.

Open follow-ups. Proposition 4.1(ii) characterises $\ker D_n$ via the unsigned vertex–edge incidence operator B_n . The natural follow-up is the structure of $\ker D_n \setminus \text{range}(H_{n+1})$ when G_n has non-trivial $\ker B_n$ (e.g. $G_n = C_4$), and whether such “incidence-cocycle” kernel directions form an invariant sector under the cascade-mechanism flow $e^{-t\tilde{A}_{n+1}}$ (numerically they do not in general, since $\tilde{A}_{n+1}$ does not preserve the harmonic subspace). A separate question is whether a non-linear cascade-flow modification (gradient flow on a modified residual that absorbs the defect into the dynamics)

preserves the mechanism’s structure while admitting strict intertwining; this is out of scope for the present paper. The P3/P4-tower lift under (L1)–(L3) is the most load-bearing follow-up.

9 Reproducibility

All computations in §§2–6 are verified to exact rational precision (`Fraction` arithmetic) by the script `papers/cascade-refinement-compat/scripts/explore_defect.py`. A reproducer with frozen reference outputs lives at `papers/cascade-refinement-compat/repro/`; from the repository root, one invocation

```
cd papers/cascade-refinement-compat/repro
bash run_audit.sh
```

re-runs the verification on Towers I and II and diffs against the frozen expected outputs. A clean run prints `REFINEMENT COMPAT AUDIT PASSED`.

References

- [1] Lee Smart. Cascade closure: A conditional compatibility template for geometric state selection. Preprint, Institute of Vibrational Field Dynamics. Local source: `papers/cascade-mechanism/cascade-mechanism.tex`. The consumer API (O0)–(O3) discharged in part by the present paper is in `cascade-mechanism.tex` §3.4., 2026.
- [2] Lee Smart. Cascade closure dynamics: Finite closure functionals, gradient flow, and refinement compatibility. Preprint, Institute of Vibrational Field Dynamics. Local source: `papers/cascade-closure-dynamics/cascade-closure-dynamics.tex`. The `coboundary-refinement` relation $p^1 d_{n+1} = \frac{1}{2} d_n p^0$ used as the load-bearing source identity is in this paper at lines 896–937 (Proposition)., 2026.
- [3] Lee Smart. Cascade refinement spaces: Finite-level state spaces and bonding maps. Preprint, Institute of Vibrational Field Dynamics. Local source: `papers/cascade-refinement-spaces/cascade-refinement-spaces.tex`., 2026.